INCH-POUND
A-A-59322C
June 11, 2024
Superseding
A-A-59322B
October 14, 2009

COMMERCIAL ITEM DESCRIPTION
OHMMETER

The general Services Administration has authorized the use of this Commercial Item description, for all federal agencies.

1 SCOPE. This Commercial Item Description (CID) describes an Ohmmeter herein referred to as equipment. This CID is meant as a minimum requirement in which only those manufactures meeting or surpassing the following requirements are to be supplied per this CID.

2 SALIENT CHARACTERISTICS. The equipment shall be capable of operation within the accuracies, limits, and specifications herein. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment assembly shall be Class 1 as described in the classification and class paragraphs of the latest version of MIL-PRF-28800 except as specified herein.

2.1 Size. The equipment dimensions shall permit efficient portable usage. The equipment's weight, including any and all accessories (excluding written manuals) shall not exceed 20 pounds.

2.2 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein.

2.2.1 Water Resistance. The equipment shall not be required to be watertight and splash proof. Drip proof requirements shall conform to MIL-PRF-28800, Class 2 while not operating and covers are closed.

2.2.2 Explosive Atmosphere. The equipment shall meet the explosive atmosphere requirements of MIL-PRF-28800 or the requirements of MIL-STD-810, METHOD 511.4, PROCEDURE I.

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2.2.3 Dust Resistance. Dust resistance testing shall not be required.

2.2.4 Salt Exposure. Salt exposure verification testing shall not be required.

2.2.5 Temperature Operating. The equipment shall meet its performance and accuracy requirements in an operating environment of at least 0 ºC to +40 ºC.

2.2.6 Non-operating Temperature. The equipment shall meet its performance and accuracy requirements in non-operating environment of at least -20 ºC to +71 ºC.

2.2.7 Transit Drop. The transit drop requirements shall conform to MIL-PRF-28800, Class 2

2.2.8 Random Vibration. The random vibration requirements shall conform to MIL-PRF-28800, Class 2.

2.3 Reliability. Unless otherwise specified, the design of the equipment shall be such that under normal use and operation the equipment does not fail within 2000 hours of operation with a statistical certainty of 95%.

2.4 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components, which are adjustable, by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater, based on an operating time of 2000 hours per year.

2.5 Electrical Power Sources and Connections. The equipment shall operate from an internal power source. Batteries shall be provided with the equipment. Batteries shall be easily accessible for maintenance and weatherproof from the environment. The equipment shall operate in accordance with the steady state conditions, transient state conditions, and interruption of power source requirements outlined the latest version of MIL-PRF-28800.

2.6 Internal DC Power. DC power shall be supplied by rechargeable internal batteries. The equipment shall be capable of operating away from AC power source. The batteries shall have the capability of operating the unit with full performance for a period of not less than eight hours over the entire temperature operating range.

2.7 Accessories. Accessories to be supplied by the manufacturer and listed in the manufacturer's catalog or data sheet, a written tutorial/user reference manual, a traceable calibration certificate, and a complete set of test leads with provisions for storage.

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2.8 Warm-up Performance. The warm-up requirement shall be met when the equipment has been subjected to a 2-hour not operating temperature stabilization period. The equipment shall perform to specifications after a warm-up period not to exceed thirty minutes after initiating power and all calibration and self-check routines have been accomplished.

2.9 Performance Characteristics.

2.9.1 Ranges. The equipment shall have the capability to perform measurements with a minimum of 4 ranges from 0 to 2000 milliohms with sufficient range selection to meet the resolution and accuracy stated herein.

2.9.2 Accuracy. The accuracy of the equipment shall be at least ± 1% of the reading on the selected range.

2.9.3 Maximum Test Current. The maximum test current of the equipment shall not be greater than 1.5 Amperes.

2.9.4 Maximum Test Voltage. The maximum test voltage of the equipment shall not be greater than 1 volt.

2.9.5 Measurement Method. The equipment shall have a four terminal measurement method to assure measurement accuracy.

2.9.6 Display. The equipment shall have a 3.5-digit display with the resolution for the required accuracy stated herein. The display shall also provide a means for illumination for use in low light conditions.

2.9.7 Resolution. The equipment shall have a minimum of 3.5-digits of resolution.

2.9.8 Probe. This equipment shall have connection for hand Grip probes as specified in MIL-DTL-26500 or MIL-DTL-83723 for new design as stated herein.

2.9.8.1 Shell Style. The connector shall have Straight Plug shell style.

2.9.8.2 Class. The connector shall have non-conductive of Class R.

2.9.8.3 Shell Size. The connector shall have minimum shell size 14.

2.9.8.4 Coupling Type. The connector shall have Bayonet Coupling Type.

2.9.8.5 Contact Type. The connector shall have Socket Contact Type.

2.9.8.6 Alternate Keying. The connector shall have Normal Alternate Keying.

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2.9.9 Protection Caps. The connector face shall have Protection Caps.

2.9.10 Underwriters Laboratories Sticker. The equipment shall have Underwriters Laboratories (UL) per UL 913 sticker on.

3 REGULATORY REQUIREMENTS.

3.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

3.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) subcontractors comply with this requirement.

4 PRODUCT CONFORMANCE PROVISIONS.

4.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6 NOTES.

6.1 Sources of documents.

6.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained at https://assist.dla.mil.

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6.1.2 The Code of Federal Regulations (CFR) may be obtained at https://www.gpoaccess.gov/cfr/index.html or available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.1.3 FAR and DFARS may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of the FAR may be obtained from http://www.arnet.gov/far/. Electronic copies of the DFARS may be obtained from https://www.acq.osd.mil/dp/dars/dfars.html.

6.1.4 Underwriters Laboratories (UL) per UL 913. Hardcopy of UL 913 standards and technical publications are available from Global Engineering Documents, the exclusive distributor of UL 913 standards at https://global.ihs.com.

6.2 Key Words.

Internal power source
Portable
Range selection

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CONCLUDING MATERIAL

Custodian:
Air Force - 184

Preparing activity:
Air Force - 184

Agent:
Air Force – 110

Project 6625-2024-002

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